Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/18/2006
PI Name:	Shaw, Benjamin Ph.D.		
Project Title:	Combustion Experiments in Reduced Gravity with Two-Component Miscible Droplets		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bdshaw@ucdavis.edu	**Fax:**	FY 530-752-4158
PI Organization Type:	UNIVERSITY	**Phone:**	530-752-4130
Organization Name:	University of California, Davis		
PI Address 1:	Department of Mechanical and Aeronautical Engineering		
PI Address 2:			
PI Web Page:			
City:	Davis	**State:**	CA
Zip Code:	95616-5294	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	08/30/2004	**End Date:**	12/21/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04AA65A		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		
	The original objective of this research was to study the combustion of bi-component miscible droplets in reduced-gravity environments so that buoyancy effects are rendered negligible. The program was termed the Bi-Component Droplet Combustion Experiment (BCDCE). Of particular interest were the burning rate (droplet history), the time to flame contraction, potential microexplosions, Marangoni flows, and sooting behaviors. The experiments used liquid mixtures selected to display influences of large component volatility differences and liquid species diffusion on combustion phenomena. The program was both experimental and theoretical in nature. Experiments included reduced-gravity studies as well as normal-gravity comparison experiments. In fall 2005, the BCDCE program was realigned with a different program that includes a team of investigators from various universities as well as from NASA. This program, which is termed the Flammability and Extinction Experiment		

Task Description:	(FLEX), is focused upon using droplets to study flammability and extinction under reduced-gravity conditions of relevance to spaceflight and human exploration of space. The FLEX program is both experimental and theoretical in nature. Experiments include reduced-gravity studies as well as normal-gravity comparison experiments. In the research, individual droplets are dispensed, deployed and ignited in reduced gravity (as well as normal gravity). Theoretical approaches include developing computational and analytical models, with a particular emphasis on evaluating influences of gas-phase diffusive transport as well as droplet internal flows, which may arise from Marangoni stresses as well as droplet deployment and ignition. The objective of the BCDCE research was to study the combustion of bi-component miscible droplets in reduced-gravity environments so that buoyancy effects were rendered negligible. Of particular interest were the burning rate (droplet history), the time to flame contraction, potential microexplosions, Marangoni flows, and sooting behaviors. The experiments used liquid mixtures selected to display influences of large component volatility differences and liquid species diffusion on combustion phenomena. The objective of the FLEX program is to employ droplets to study droplet flammability and extinction under reduced-gravity conditions of relevance to spaceflight and human exploration of space. The research includes experimental and theoretical components. In the experiments, individual droplets are dispensed, deployed and ignited in reduced gravity as well as normal gravity. Theoretical approaches include developing computational and analytical models of flammability and extinction, with a particular emphasis on evaluating influences of gas-phase diffusive transport as well as droplet internal flows, which may arise from Marangoni stresses as well as droplet deployment and ignition. The BCDCE research extended previous research on combustion of pure liquid fuels to mixtures of pure fuels, which more closely approximate practical fuels. The results from this research program were relevant to design of practical combustion systems that employ liquid fuels such as engines and oil-fired burners. The FLEX results will be useful for improving understanding of combustion and fire-related phenomena under reduced gravity conditions, e.g., influences of diffusive transport and droplet internal flows on flammability and extinction.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The BCDCE research extended previous research on combustion of pure liquid fuels to mixtures of pure fuels, which more closely approximate practical fuels. The results from this research program were relevant to design of practical combustion systems that employ liquid fuels such as engines and oil-fired burners. The FLEX results will be useful for improving understanding of combustion and fire-related phenomena under reduced gravity conditions, e.g., influences of diffusive transport and droplet internal flows on flammability and extinction.
Task Progress:	Work has principally been in the following areas: reduced-gravity and normal-gravity experiments; development of a computational model; and development and use of the NASA Tracker Station for analysis of 16mm cine films of droplet combustion experiments. A summary of the progress is given below. Reduced-Gravity and Normal-Gravity Experiments: Reduced gravity and normal gravity combustion experiments were performed with individual fiber-supported droplets that were initially about 1 mm in diameter. Most experiments were performed with propanol, but some data were also obtained with heptane and methanol. The environment was composed of an air-inert mixture at 0.1 MPa and about 298 K, where helium, carbon dioxide and xenon were separately used as inerts. In the experiments, the amount of inert gas in the environment was progressively increased until droplets could no longer be ignited. Droplet ignitability depended on the gravitational level as well as the inert gas in the environment. In general, the amount of inert gas required to suppress combustion was higher in reduced gravity than normal gravity. Use of xenon required significantly higher mole fractions for suppression of combustion relative to helium or carbon dioxide. The data indicate that ambient gas transport properties such as Soret transport play an important role in determining flammability of droplets. Computational Model Development: Results for three-dimensional numerical calculations of vaporization histories of 25-100 micron methanol droplets supported on thin fibers were developed. These droplets were assumed to be spherical. Calculations were carried out for high temperature and high pressure external flow conditions, and droplets vaporized in air. Fiber diameters of 1/20 to 1/5 of the initial droplet diameters were selected to enable investigation of the influences of support fibers on the droplet behavior. A unique feature of these calculations is that effects of thermocapillary flows were considered. The effects of the fiber on droplet internal and external circulation at low and high Reynolds numbers were investigated. A thick fiber increases the net heat transfer rates to the droplet. Supported and unsupported droplets vaporize at similar rates at high Reynolds numbers. At lower Reynolds numbers, unsupported droplets vaporize faster. Further three-dimensional numerical calculations involved modeling isolated 100 micron methanol droplets burning in microgravity environments at 1500 K and 50 atm [4]. Three flow configurations were considered: (1) a spherical unsupported droplet; (2) a droplet fully supported in the middle of a 10 micron quartz fiber with a solid-liquid contact angle of 30 degrees; and (3) a droplet suspended at the end of a 10 micron quartz fiber such that the droplet climbs up the fiber with a contact angle of 60 degrees. Cases (2) and (3) involved non-spherical droplets with droplet shapes determined by the liquid-fiber contact angle. Marangoni convection due to thermocapillary stresses was suppressed due to limitations of the speed of present-day desktop computers. Water condensation on the droplet surface was also suppressed. Ignition occurs the earliest in the suspended droplet, followed by the unsupported droplet and lastly for a fully supported droplet. It was found flame shapes and droplet heat transfer rates were different for the three conditions investigated. Use of the NASA Tracker Station: The NASA Tracker Station is being used at UC Davis for capture and analysis of 16 mm films from droplet combustion experiments. A high resolution Canon EOS 20D digital camera and a Canon MP-E macro lens are used in conjunction with the Tracker film transport system to form an automated system where frames of film are converted into digital images. The camera resolution is sufficient to allow individual film grains to be imaged, thus providing very high resolution images of droplets and flames. The Tracker system is controlled via a computer. The MATLAB technical computing software environment along with the Image Processing Toolbox is used to track droplet characteristics during combustion. The coordinates of droplet outlines are obtained and parameters of interest such as droplet diameter are calculated to plot time histories. The analysis method produces high resolution and accurate results in an efficient

	manner. Other: A paper describing theory to predict influences of thermal pyrolysis of fuel on soot shell locations was accepted by a journal. A paper describing experiments related to influences of pressure on reduced-gravity combustion of propanol droplets was accepted by a journal.
Bibliography Type:	Description: (Last Updated: 04/29/2010)
Abstracts for Journals and Proceedings	Shringi D, Dwyer HA, Shaw BD. "Numerical Study of Fiber Influences on Droplet Flow and Visualization." 20th International Colloquium on the Dynamics of Explosions and Reactive Systems, Montreal, Canada, July 31 - August 5, 2005. 20th International Colloquium on the Dynamics of Explosions and Reactive Systems, July 2005, Abstract 168. , Jul-2005
Articles in Peer-reviewed Journals	Shaw BD, Dakka SM. "Theory of influence of fuel pyrolysis on soot-shell standoff ratios in reduced-gravity droplet combustion." Combust Sci Tech. 2005 Oct;177(10):1939-59. http://dx.doi.org/10.1080/00102200590970339 , Oct-2005
Articles in Peer-reviewed Journals	Dakka SM, Shaw BD. "Influences of Pressure on Reduced-Gravity Combustion of Propanol Droplets." Microgravity Sci Technol. (In Press), March 2006. , Mar-2006
Articles in Peer-reviewed Journals	Shaw BD. "Theory of spherically symmetrical droplet combustion with gas-phase fuel pyrolysis." Combust Sci Tech. 2005 May-Jun;177(5-6):1151-66. http://dx.doi.org/10.1080/00102200590927021 , May-2005
Articles in Peer-reviewed Journals	Shaw BD, Dee V. "Influence of the gas phase inert on reduced-gravity combustion of decane/hexadecane droplets." Microgravity Sci Technol. 2005; 17(1):26-34. , May-2005
Papers from Meeting Proceedings	Shringi D, Dwyer HA, Shaw BD. "Supporting Rod Influences on Methanol Droplet Combustion." 2006 Spring Technical Meeting of the Western States Section of the Combustion Institute, Boise State University, Boise, ID, March 27-28, 2006. 2006 Spring Technical Meeting of the Western States Section of the Combustion Institute, March 2006, Paper 06S-04. , Mar-2006
Papers from Meeting Proceedings	Wei JB, Shaw BD. "Reduced Gravity Combustion of Propanol Droplets with Various Inert Gases." 2005 Fall Meeting of the Western States Section of the Combustion Institute, Stanford University, Stanford, CA, October 17-18, 2005. 2005 Fall Meeting of the Western States Section of the Combustion Institute, October 2005, Paper 05F-54. , Oct-2005
Papers from Meeting Proceedings	Shringi D, Dwyer HA, Shaw BD. "Effects of Supporting Rods on Fuel Droplet Heating and Vaporization with Surface Tension Effects." 2005 Fall Meeting of the Western States Section of the Combustion Institute, Stanford University, Stanford, CA, October 17-18, 2005. 2005 Fall Meeting of the Western States Section of the Combustion Institute, October 2005, Paper 05F-53. , Oct-2005